

Integrability of Subsampled Compositional Diffusions

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Abstract

Compositional diffusion inference combines many factor scores, making subsampling attractive when evaluating every factor is expensive. For a Feynman–Kac correction, however, an unbiased estimate of the instantaneous potential is exponentiated to obtain a positive particle weight. We study whether the resulting population operator has a finite normalizing constant. For Gaussian factors sampled with replacement, we give an exact reachable-variance recursion requiring only two extreme batches per step, independently of every finite batch size. A four-factor example has a finite full-factor update and an infinite subsampled normalizer for every finite batch size, even as local estimator mean squared error tends to zero. For uniform sampling without replacement, an exact prefix/suffix recursion instead depends on batch size. We also construct an unbiased control-variate estimator that preserves the deterministic quadratic tail of equal-component-variance Gaussian mixtures. Its strict integrability domain agrees with that of the full-factor discretization. Oracle-score and learned-mixture experiments separate population integrability, finite-particle behavior, and learning error; finite tail control need not improve sampling accuracy. The results identify tail curvature as a validity condition for exponential subsampling, alongside variance and computational cost.

1 Introduction

Compositional inference uses separately available conditional distributions to build a posterior for multiple observations or groups. This structure is useful in simulation-based inference, where amortized factor models can be reused across datasets. Recent work develops diffusion posterior sampling for tall data (Linhart et al., 2026), and compositional amortized inference for large hierarchical models (Arruda et al., 2026). Evaluating all factor scores at each particle and diffusion time can nevertheless dominate inference cost. A natural computational question is whether a small random batch of factor scores can replace the full sum.

That question becomes delicate when composition is accompanied by Feynman–Kac weighting. The correction contains a quadratic expression in the composed score. An unbiased estimator of this expression is straightforward to construct with independent, uniformly sampled factor indices. Its exponential is positive, and any finite collection of sampled states produces finite real-valued log weights in exact arithmetic. Neither property guarantees that the population weight can be integrated. A rare batch can create an excessively positive quadratic potential in a Gaussian tail, making its expected exponential infinite.

This paper studies that existence question for a specified weighted Euler discretization. The issue is separate from the established mismatch between composed intermediate distributions and forward-noised target distributions, and from learned score error (Linhart et al., 2026; Soiffer et al., 2026). Oracle experiments first isolate the random-weight effect. A controlled conditional-density learning study then uses exact noised scores of learned mixtures, allowing posterior learning error and particle error to be measured separately. We also distinguish finite normalizing constants from small endpoint error: a valid discretization may still be inaccurate, and a finite-particle sample may appear accurate when its population operator is undefined.

Our main result makes this distinction computable. For scalar Gaussian factors with non-negative residual precisions, the largest variance reachable over all finite batch histories evolves by a two-choice recursion. The choices are batches containing only the weakest or only the strongest factor. These events have positive probability for every finite with-replacement batch size, so increasing the batch size can reduce estimator mean squared error without changing the integrability certificate. A closed-form four-factor example exhibits this separation in a single step. Uniform sampling without replacement instead admits a batch-size-dependent prefix/suffix recursion. For non-Gaussian factors with a shared variance within each Gaussian mixture, an affine control preserves the deterministic quadratic tail and the full-factor strict integrability domain. The empirical study reports both the certificates and the observed particle errors.

2 Compositional flow and random exponential weights

Let φ be the standard Gaussian density on $\mathbb{R}^d$. Factor density $q_{g,0}$ evolves under the variance-preserving Ornstein–Uhlenbeck process

$$dX_u = -\frac{1}{2}X_u du + dW_u, \quad X_u = e^{-u/2}X_0 + \sqrt{1 - e^{-u}}\xi. \quad (1)$$

Write $s_g(x, u) = \nabla \log q_{g,u}(x)$, $r_g = s_g + x$, and $R = \sum_{g=1}^G r_g$. The unnormalized compositional path is

$$\rho_u(x) = \varphi(x)^{1-G} \prod_{g=1}^G q_{g,u}(x), \quad \nabla \log \rho_u = -x + R. \quad (2)$$

We assume this path is integrable over the interval considered. With reverse time $t = U - u$, define

$$b_\kappa(x, u) = \frac{1 + \kappa}{2}R(x, u) - \frac{\kappa}{2}x, \quad g(x, u) = \frac{1}{2} \left(\|R(x, u)\|^2 - \sum_{g=1}^G \|r_g(x, u)\|^2 \right), \quad (3)$$

where $\kappa > 0$. Direct differentiation gives

$$-\partial_u \rho_u = -\nabla \cdot (b_\kappa \rho_u) + \frac{\kappa}{2} \Delta \rho_u + g \rho_u. \quad (4)$$

Thus the normalized continuous flow is represented by a Feynman–Kac diffusion whenever its required moments exist. Appendix A derives this identity. Our results concern the following discrete operator; they do not assert that its finite-step target equals ρ_0 .

For a step $h > 0$ at level u , independently draw a batch $B = (I_1, \dots, I_M)$, with each index uniform on $\{1, \dots, G\}$. For $M \geq 2$, define

$$\hat{R}_B = \frac{G}{M} \sum_{i=1}^M r_{I_i}, \quad (5)$$

$$\hat{g}_{U,B} = \frac{1}{2} \left\{ \frac{G^2}{M(M-1)} \sum_{i \neq j} r_{I_i}^\top r_{I_j} - \frac{G}{M} \sum_i \|r_{I_i}\|^2 \right\}. \quad (6)$$

Both are conditionally unbiased at every fixed (x, u) . The plug-in estimator $\hat{g}_N = \frac{1}{2}(\|\hat{R}\|^2 - \frac{G}{M} \sum_i \|r_{I_i}\|^2)$ is also considered. Its conditional bias is $G^2 \text{tr Cov}_I(r_I)/(2M)$.

For either estimator, let $\hat{b}_B = (1 + \kappa)\hat{R}_B/2 - \kappa x/2$. The positive kernel and its normalized update are

$$Q_{u,h}f(x) = \mathbb{E}_{B,\xi} \left[e^{h\hat{g}_B(x,u)} f\left(x + h\hat{b}_B(x,u) + \sqrt{\kappa h}\xi\right) \right], \quad \Phi_{u,h}(\eta)(f) = \frac{\eta Q_{u,h}f}{\eta Q_{u,h}1}. \quad (7)$$

The denominator must belong to $(0, \infty)$. Conditional unbiasedness of $\hat{g}$ does not imply $\mathbb{E}_B e^{h\hat{g}} = e^{hg}$ or that $\eta Q1$ is finite. We analyze this denominator before making any particle-convergence claim.

3 Gaussian integrability certificates

In one dimension, Gaussian factors have $r_g(x, u) = -a_g(u)x + d_g(u)$. At a fixed level, suppress u and suppose $a_g \geq 0$. For a batch, put $m = M^{-1} \sum_i a_{I_i}$ and $s^2 = M^{-1} \sum_i (a_{I_i} - m)^2$. The drift and potential have the form

$$\hat{b}_B(x) = -\frac{1}{2}((1 + \kappa)\hat{A}_B + \kappa)x + \text{constant}, \quad \hat{g}_B(x) = \hat{c}_B x^2 + \text{linear terms}, \quad (8)$$

where $\hat{A}_B = Gm$, and

$$\hat{c}_{U,B} = \frac{1}{2} \left[G(G-1)m^2 - \left(\frac{G^2}{M-1} + G \right) s^2 \right], \quad \hat{c}_{N,B} = \frac{1}{2} [G(G-1)m^2 - Gs^2]. \quad (9)$$

For the full-factor update, $A = \sum_g a_g$ and $c = (A^2 - \sum_g a_g^2)/2$.

Lemma 1 (Conditional Gaussian update). Suppose the normalized state conditional on a batch history is Gaussian with variance $V > 0$. Its next conditional normalizer is finite if and only if

$$D_B(V) = 1 - 2h\hat{c}_B V > 0. \quad (10)$$

When finite, the propagated conditional law is Gaussian with variance

$$F_B(V) = \frac{[1 - h((1 + \kappa)\hat{A}_B + \kappa)/2]^2 V}{D_B(V)} + \kappa h. \quad (11)$$

The criterion and variance are independent of the Gaussian mean and the linear terms in the potential.

Proof. Multiplication by the weight changes the Gaussian precision from V^{-1} to $V^{-1} - 2h\hat{c}_B$. A nonpositive precision gives an infinite integral on $\mathbb{R}$, including the zero-precision case. For positive precision, completing the square gives tilted variance $V/D_B(V)$. The affine Euler map and independent Gaussian noise give (11). $\square$

Every batch history has positive probability. If one of its conditional normalizers is infinite, Tonelli's theorem makes the joint nonnegative integral infinite. If all histories remain finite for a finite number of steps, their finite sum is finite. A direct check appears to require exponentially many histories. The following result removes that enumeration.

Theorem 2 (Two extreme batches suffice). Fix a finite time grid, $G \geq 2$, $M \geq 2$, and $\kappa > 0$. Assume scalar Gaussian factors with $a_g(u_k) \geq 0$ and an initial Gaussian variance $V_0 > 0$. For either estimator in (9), set $V_{\max,0} = V_0$. At step k , let $a_- = \min_g a_g(u_k)$, $a_+ = \max_g a_g(u_k)$ and define, for $z \in \{a_-, a_+\}$,

$$D_z = 1 - h_k G(G-1)z^2 V_{\max,k}, \quad F_z = \frac{[1 - h_k((1 + \kappa)Gz + \kappa)/2]^2 V_{\max,k}}{D_z} + \kappa h_k. \quad (12)$$

If some $D_z \leq 0$, the population normalizer is infinite at this step. Otherwise set $V_{\max,k+1} = \max(F_{a_-}, F_{a_+})$ and continue. Until failure, this is exactly the largest reachable conditional variance. Consequently the integrability decision is identical for every finite $M \geq 2$, for both the plug-in and unbiased estimators.

Proof. Both coefficients in (9) are at most $G(G-1)m^2/2$, with equality for a constant batch. Since $0 \leq m \leq a_+$, a nonpositive denominator can occur if and only if it occurs for the constant strongest-factor batch at the largest reachable variance. On the common positive-denominator domain, $F_B(V)$ is nondecreasing in V and in $\hat{c}_B$ with $\hat{A}_B$ fixed. It remains to maximize

$$f(m) = \frac{V[1 - h((1 + \kappa)Gm + \kappa)/2]^2}{1 - hVG(G-1)m^2}, \quad m \in [a_-, a_+].$$

This function is convex. Indeed, with affine $n(m) = \sqrt{V}[1 - h((1 + \kappa)Gm + \kappa)/2]$ and positive concave $d(m) = 1 - hVG(G-1)m^2$,

$$f(m) = \sup_{z \in \mathbb{R}} \{2zn(m) - z^2 d(m)\},$$

the supremum of convex quadratics. Its maximum on the interval occurs at an endpoint. The corresponding constant batches exist for every M and have positive probability. Induction establishes attainability of each maximum and the stated decision. $\square$

The cost is $O(KG)$ to obtain the extrema and $O(K)$ scalar transitions, with no enumeration over G^M batches. For diagonal product factors, apply the recursion to each coordinate: a single failing coordinate suffices for divergence, while positivity for every coordinate certifies every history.

Uniform sampling without replacement. For a uniform subset S of size $2 \leq M \leq G$, the unbiased potential is $G(G-1) \sum_{i < j, i, j \in S} r_i^T r_j / [M(M-1)]$. Its exact Gaussian certificate depends on M . After sorting $a_1 \leq \dots \leq a_G$, only the $M+1$ subsets containing the j weakest and $M-j$ strongest factors, $0 \leq j \leq M$, need be checked. Fixing all but one selected strength makes the drift and quadratic potential coefficients affine in the remaining strength; convexity of the squared-affine-over-positive-affine variance map permits an endpoint exchange. Repeated exchanges give these prefix/suffix subsets. Appendix E gives the full theorem and multi-step proof. At $M = G$, this update is the full-factor operator.

3.1 An exact four-factor separation

Take zero-mean Gaussian factor precisions $1 + \lambda_g$ with $\lambda = (1/10, 1/5, 2, 4)$. At $u = \log 2$, the residual precisions are $a = (1/21, 1/11, 1/2, 2/3)$. The normalized compositional density is $\mathcal{N}(0, V)$ with $V = 154/355$. For $h = 1/2$ and $\kappa = 1$,

$$A = \frac{201}{154}, \quad c = \frac{346}{693}, \quad D_{\text{full}} = \frac{2503}{3195} > 0, \quad D_{\text{all strongest}} = -\frac{167}{1065} < 0. \quad (13)$$

The all-strongest batch occurs with probability 4^{-M} and has $\hat{c} = 8/3$ for either estimator. Thus the subsampled normalizer is infinite for every finite M , whereas the full-factor update has log normalizer $-\frac{1}{2} \log(2503/3195)$ and output variance

$$[1 - \frac{1}{2}(A + \frac{1}{2})]^2 \frac{154/355}{2503/3195} + \frac{1}{2}. \quad (14)$$

At each fixed state, both unbiased estimators converge in mean square as $M \rightarrow \infty$, because the index distribution has finite support. Their state-averaged mean squared errors also vanish for this Gaussian input. This limit does not interchange with the nonnegative exponential integral. For P independent particle batches, the probability of observing at least one all-strongest batch is $1 - (1 - 4^{-M})^P$; observing such a batch still does not guarantee an extreme sampled state.

4 Preserving the deterministic quadratic tail

Suppose each scalar factor is a finite Gaussian mixture whose components share variance v_g but may have different means. After forward noising, its residual score admits

$$r_g(x, u) = r_g^0(x, u) + e_g(x, u), \quad r_g^0 = -a_g(u)x + d_g(u), \quad \sup_x |e_g(x, u)| < \infty, \quad (15)$$

where $a_g(u) = [1 - e^{-u} + e^{-u}v_g]^{-1} - 1$. One may use $d_g = e^{-u/2}\bar{\mu}_g/[1 - e^{-u} + e^{-u}v_g]$, with $\bar{\mu}_g$ the mixture mean. The bound follows because the posterior component mean lies in the convex hull of finitely many component means. A Gaussian fit using the total mixture variance generally does not satisfy this bounded-residual property.

Write $R_0 = \sum_g r_g^0$ and $Q_0 = \sum_g \|r_g^0\|^2$. Compute both affine/quadratic functions exactly. The tail-control estimator is

$$\begin{aligned} \hat{R}_T &= R_0 + \frac{G}{M} \sum_i e_{I_i}, \\ \hat{G}_T &= \frac{1}{2}(\|R_0\|^2 - Q_0) + R_0^\top \frac{G}{M} \sum_i e_{I_i} - \frac{G}{M} \sum_i (r_{I_i}^0)^\top e_{I_i} \\ &\quad + \frac{1}{2} \left\{ \frac{G^2}{M(M-1)} \sum_{i \neq j} e_{I_i}^\top e_{I_j} - \frac{G}{M} \sum_i \|e_{I_i}\|^2 \right\}. \end{aligned} \tag{16}$$

Taking conditional expectations proves unbiasedness. Expanding $r_g = r_g^0 + e_g$ gives exactly the full potential. Moreover, every batch has the same quadratic coefficient c and the same linear drift coefficient as the full-factor construction; all random contributions to the potential grow at most linearly.

Theorem 3 (Strict integrability domain of tail control). Consider scalar equal-component-variance Gaussian mixture factors, a finite grid with $\kappa > 0$, and an initial density satisfying

$$\log p_0(x) = -x^2/(2V_0) + O(1 + |x|). \tag{17}$$

At each step propagate the full-factor quadratic variance using (11) with (A, c) . If all resulting denominators are strictly positive, both the full-factor and tail-control population normalizers are finite. If the first nonpositive denominator is strictly negative, both normalizers are infinite there. This statement holds for every finite $M \geq 2$. No claim is made at a zero denominator for general mixtures.

Proof. For a density with (17) and a potential $cx^2 + O(1 + |x|)$, weighting replaces its quadratic variance by $V/(1 - 2hcV)$ when the denominator is positive; a negative denominator forces an infinite integral. An affine map plus a bounded displacement followed by independent Gaussian noise preserves the quadratic-tail form, with variance (11); Appendix B proves this step. Equation (16) makes these quadratic coefficients common to every batch. Induction over the finite histories gives the result. $\square$

The theorem certifies existence of the discretized population flow. It does not remove Euler error or the effect of exponentiating the bounded residual estimator. Its computational requirement is also explicit: the affine coefficient sums can be prepared once per time level in $O(Gd)$ work, followed by M nonlinear factor evaluations per particle. The control requires known component tail variances, which limits its direct applicability to arbitrary learned score models.

More generally, an arbitrary affine control also admits an exact strict-domain certificate: it suffices to evaluate the G constant batches, each containing one repeated factor. Appendix C proves this extension and shows that the paired correction has the same strict integrability domain. Thus the moment-matched and tail-matched controls in our experiments can be classified using the same test, although only the latter preserves the full-factor domain by construction.

5 Experiments

The fixed baseline protocol comprises 1,260 main cells and 720 supplementary cells. It uses exact Gaussian and Gaussian-mixture factor scores, isolating random-weight effects from learned-model approximation. Baseline benchmark cells use 8,192 particles, $M = 4$, $G \in \{16, 64\}$, $d \in \{1, 8\}$, $K \in \{128, 512, 2048\}$, and five random seeds. The reverse-time grid is $u_k = 20(1 - k/K)^2$, the initial distribution is $\mathcal{N}(0, I)$, and $\kappa = 1$. The finite- U initialization is an approximation to the exact compositional path. Reference marginals are computed by numerical quadrature on 65,537 points in $[-12, 12]$.

The main comparison includes full-factor weighting, an unweighted composed drift, plug-in weighting, unbiased quadratic weighting, a paired cumulant correction, and two moment-matched Gaussian control variants. The supplementary benchmark evaluates tail control

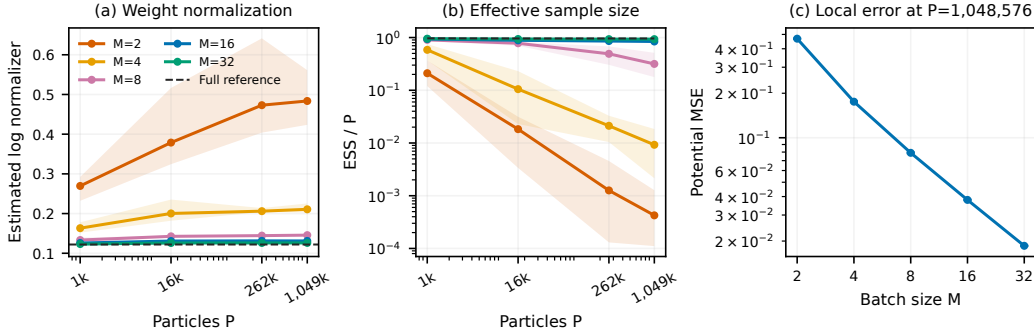


Figure 1: One-step Gaussian counterexample. Panels (a,b) show medians and interquartile ranges over 20 seeds. The dashed reference is the exact full-factor log normalizer in (a) and the empirical full-factor ESS fraction in (b). Panel (c) shows mean potential MSE with one standard deviation at $P = 1,048,576$. Every subsampled curve has an infinite population normalizer.

and its paired correction on both mixture families. Paired corrections use two independent batches and the positive weight $\exp\{h(\hat{g}_1 + \hat{g}_2)/2 - h^2(\hat{g}_1 - \hat{g}_2)^2/8\}$, with the averaged drift; they therefore require twice as many sampled factor evaluations. This correction is an empirical ablation, with no claim of exact exponential unbiasedness.

For the one-step example in (13), the supplementary study uses $P \in \{1,024, 16,384, 262,144, 1,048,576\}$, $M \in \{2, 4, 8, 16, 32\}$, and 20 seeds, together with the full-factor update. We retain histograms and the 1,024 largest normalized particle weights for each one-step cell. The benchmark retains complete particle arrays and reference distributions.

All 1,980 baseline cells completed on SCNet RTX 3080 GPUs. We verified the source versions, 13,506 recovered file hashes, all 40 task exit records, and the complete cell configuration lists. Benchmark metrics were recomputed using independent numerical routines. Adaptive quadrature over the real line also verified all 54 reference coordinates, including analytic Gaussian checks. A separately frozen extension adds 1,080 oracle sampling cells and 1,000 learned-density sampling cells, with five complete network trainings (Appendix I).

5.1 Local error decreases while the normalizer remains infinite

Figure 1 shows the one-step experiment. At $P = 1,048,576$, increasing M from 2 to 32 reduces mean potential MSE from 0.46798 to 0.01857, approximately a factor of 25. The median ESS fraction rises from 0.000424 to 0.932706. Nevertheless, (13) gives an infinite population normalizer for both batch sizes and every intermediate finite M .

For $M = 32$, the empirical mean log normalizer is 0.126047 with standard deviation 0.000259 across 20 seeds. The full-factor result is 0.122030 ± 0.000234 , close to its exact value 0.122049. No nonintegrable batch was sampled in these 20 runs of 1,048,576 particles with $M = 32$. The all-strongest event alone has probability $4^{-32} \approx 5.42 \times 10^{-20}$ per batch. Its absence in this finite experiment cannot establish existence of the population target. For $M = 2$, the median estimated log normalizer rises from 0.270 at $P = 1,024$ to 0.484 at $P = 1,048,576$, accompanied by severe ESS loss. The proof supplies the divergence result; these finite observations illustrate why it can be visible at small M and inconspicuous at larger M .

5.2 Certificates across the benchmark grid

Appendix Table 2 reports validity over the 12 configurations per family, formed by two group counts, two dimensions, and three time grids. Seeds do not change a certificate. Raw plug-in, unbiased, and paired estimators fail all Gaussian and ordinary-mixture configurations. Moment matching makes the Gaussian estimator exact, but does not certify any ordinary-

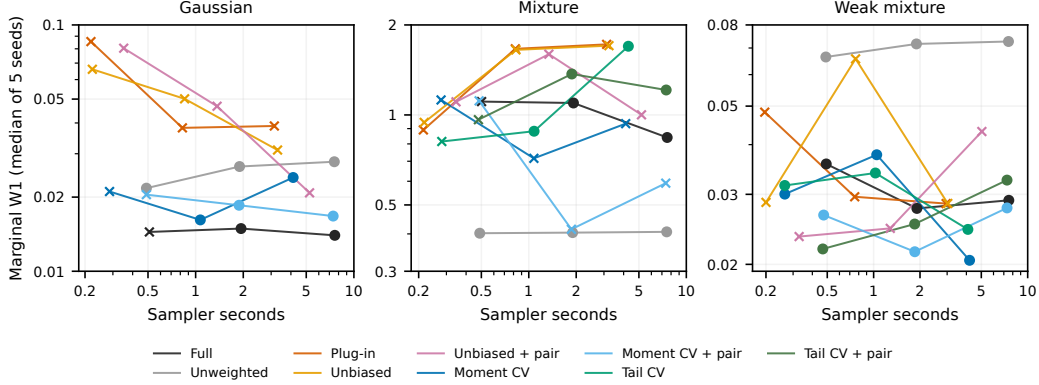


Figure 2: Observed error versus recorded sampler time at $G = 64, d = 8$, with $K \in \{128, 512, 2048\}$. Each point is a median over five seeds. Circles denote finite population normalizers and crosses denote infinite ones. Lines connect a method’s three time grids and are not fitted performance models.

mixture configuration. Tail matching recovers the full-factor strict domain in both mixture families. The complete configuration-level grid is provided in Appendix F.

The full-factor discretization also fails for $G = 64, K = 128$ in the Gaussian and ordinary-mixture families. Its Gaussian denominator first becomes negative at step 100, reaching approximately -0.003111 . Yet the observed mean marginal W1 is only 0.0141 in one dimension and 0.0176 in eight dimensions. Thus a small endpoint sample discrepancy can coexist with an undefined population update even before subsampling. For finite Gaussian configurations, exact weighted-Euler moment recursion separates time error from particle error: at $G = 64, K = 2048$, the population W1 is approximately 0.000493, while the eight-dimensional finite-particle mean marginal W1 is 0.0149.

5.3 A finite domain does not guarantee accurate or faster inference

Figure 2 and Table 3 compare actual sampler time and endpoint error. For the difficult eight-dimensional ordinary mixture with $G = 64, K = 512$, all weighted methods have substantial sampling error. Full-factor W1 is 0.881 ± 0.372 , tail control is 1.004 ± 0.327 , and paired tail control is 1.358 ± 0.188 . Their mean total-variation errors over the 256 coordinate-sign patterns are respectively 0.961, 0.975, and 0.998, confirming poor joint-mode coverage. The paired moment control has lower marginal W1, 0.456 ± 0.228 , despite an infinite population normalizer. These outcomes demonstrate that finite integrability, marginal accuracy, and joint coverage are distinct properties.

On the weak mixture, paired tail control yields W1 0.025 ± 0.005 , compared with 0.028 ± 0.007 for full weighting, at nearly the same recorded sampler time. Unpaired tail control uses 16 times fewer nonlinear particle-factor evaluations and about 1.86 times less sampler time, but its W1 rises to 0.043 ± 0.026 . We therefore make no general speed-accuracy superiority claim. Gaussian full and exact-control results, complete seed-level metrics, and additional joint diagnostics are retained in the artifacts.

5.4 Without replacement: batch size changes the finite domain

The prefix/suffix certificate permits an exhaustive scan over batch sizes without enumerating subsets. Table 1 reports the smallest finite batch size among all $2 \leq M \leq G$ for the benchmark tails. This separate deterministic calculation comprises 936 configurations and agrees with direct subset enumeration in nine smaller cross-checks. The ordinary mixture shares the Gaussian component variances, and dimensions one and eight give identical thresholds here. All larger batch sizes are finite in each displayed case with a finite threshold; this observed ordering is not claimed as a general monotonicity theorem. At $G = 64, K = 512$,

Table 1: Smallest certified finite without-replacement batch size, with $U = 20$, $\kappa = 1$ and initial variance one. A dash means no finite batch exists, including $M = G$. The mixture rows use the strict tail certificate.

Tail family	G	$K = 128$	$K = 512$	$K = 2048$
Gaussian / ordinary mixture	16	13	12	12
Gaussian / ordinary mixture	64	–	60	59
Weak mixture	16	3	2	2
Weak mixture	64	22	18	17

switching from replacement to a uniform subset of size four remains below either family’s threshold. Thus removing repeated indices changes the certificate but does not by itself make a small batch valid.

5.5 Learned conditional densities with explicit tail structure

We train five mixture density networks on joint samples from a two-mode simulator, using conditional negative log likelihood and no exact-posterior labels. Each network completes 40,000 updates. Its output is a two-component Gaussian mixture with a shared, bounded component variance, so the noised score is exact conditional on the learned parameters. This architecture makes tail curvature available to the certificate. Five dataset seeds crossed with the five checkpoints give 25 pairs for each (G, d) configuration. We compare particles with both the learned composition and the true posterior, using $P = 32768$, $M = 4$ and $K \in \{512, 2048\}$; Appendix I gives the simulator, training checks and complete protocol.

All 1,000 learned sampling cells pass independent raw-sample, reference-density and certificate audits. Full weighting and both tail controls have finite population normalizers in all 600 corresponding cells; both raw unbiased estimators have infinite normalizers in all 400 remaining cells. Without replacement therefore does not restore validity at this batch size. Across the four (G, d) settings, mean W1 between the true and learned compositions ranges from 0.00536 to 0.00801, and mean KL from 0.00075 to 0.00304.

Figure 3 again separates validity from accuracy. At $G = 64, d = 8, K = 2048$, W1 against the learned composition is 0.0189 ± 0.0073 for full weighting, 0.0448 ± 0.0130 for raw without-replacement weighting, and 0.0907 ± 0.0136 for tail control without replacement. Against the true posterior, the corresponding values are 0.0201, 0.0462 and 0.0912. Tail control restores the finite domain but has larger errors in this comparison. Its recorded mean sampler time is 5.73 seconds, versus 26.09 seconds for full weighting; at $G = 16, d = 1$, it instead takes 3.86 seconds versus 2.27 seconds. These timings include subset selection and exclude separately recorded one-time neural parameter prediction. A frozen replication on 20 new datasets confirms the larger tail-control error; Appendix K also tests validity thresholds and imperfect tail controls.

6 Related work

Feynman–Kac correctors derive weighted diffusions and resampling schemes for products of pretrained marginals (Skreta et al., 2025); energy-based diffusion models also support composition through SMC (Thornton et al., 2025). Tall-data and hierarchical inference motivate score composition (Linhart et al., 2026; Arruda et al., 2026), while learned-score error can impair compositional generation (Soiffer et al., 2026). These constructions are background here. Our question concerns the finite-step population normalizer after independently subsampling the factors in both drift and exponential potential.

Random-weight particle filters explicitly use stochastic weights (Fearnhead et al., 2010); unbounded-weight convergence and particle stability require moment or flow conditions (Mbalawata & Särkkä, 2016; Whiteley, 2013). Subsampling SMC uses approximately unbiased likelihood estimators and auxiliary-variable updates (Gunawan et al., 2020). Our contributions are the exact extreme-batch and prefix/suffix certificates for (7), their differing

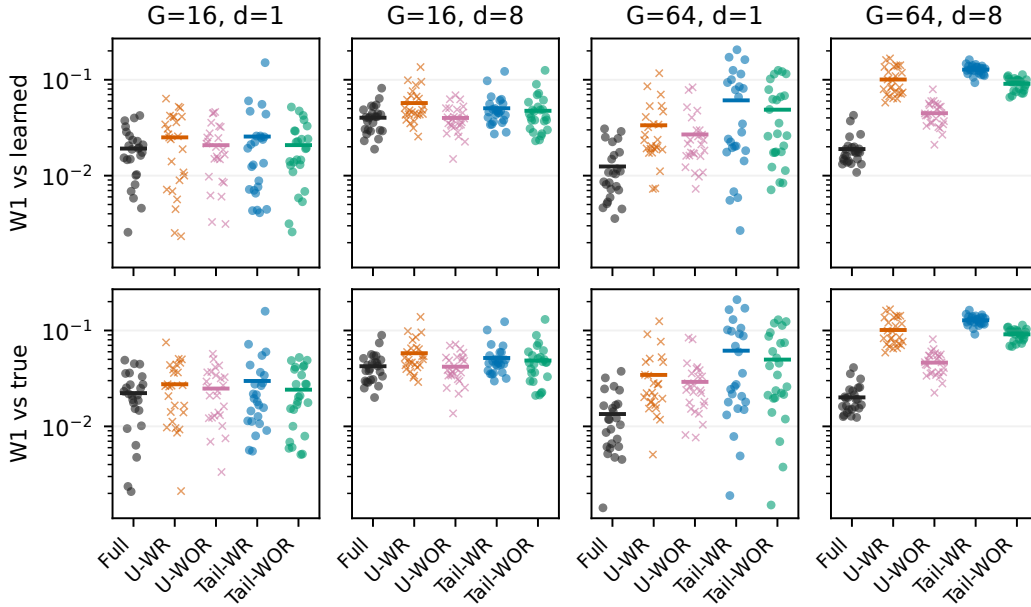


Figure 3: Learned-posterior sampling at the preselected $K = 2048$. Each panel shows all 25 crossed checkpoint/dataset pairs; horizontal bars are means. Circles denote finite population normalizers and crosses infinite ones. U is raw unbiased weighting; WR and WOR denote sampling with and without replacement. The upper row measures particle W1 against the learned composition and the lower row against the true posterior. These 25 pairs are not independent training replicates.

batch-size dependence, and preservation of the full-factor strict domain through deterministic tail curvature. The Gaussian integral and unbiased quadratic estimator are classical calculations. These certificates concern the stated Euler operator, not every Feynman–Kac simulation scheme.

7 Discussion

The certificate distinguishes three questions that are often evaluated together: existence of a population target, accuracy of its time discretization, and behavior of a finite particle approximation. With-replacement subsampling can fail the first question while producing finite samples and decreasing local mean squared error. Without replacement, batch size changes the certificate, and all subset sizes in our four-factor one-step example are finite. Tail control preserves the full-factor strict domain for the stated mixture class, but accuracy and wall-clock cost remain empirical questions. The learned experiment requires a density architecture with known Gaussian tails; it is not a certificate for an arbitrary neural score. The analysis is restricted to scalar or diagonal product tails and a fixed weighted Euler construction. Noncommuting covariances, adaptive batches and learned models beyond this tail class require additional arguments.

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A Derivation of the compositional Feynman–Kac identity

For a smooth positive factor density satisfying the Ornstein–Uhlenbeck forward equation,

$$\partial_u \log q_g = \frac{1}{2} \nabla \cdot r_g - \frac{1}{2} x^\top r_g + \frac{1}{2} \|r_g\|^2.$$

The prior φ is stationary. Consequently, with $Q = \sum_g \|r_g\|^2$,

$$-\partial_u \log \rho = -\frac{1}{2} \nabla \cdot R + \frac{1}{2} x^\top R - \frac{1}{2} Q.$$

Put $S = \nabla \log \rho = -x + R$. Since $b_\kappa = R/2 + \kappa S/2$,

$$\begin{aligned} \rho^{-1} [-\nabla \cdot (b_\kappa \rho) + \frac{\kappa}{2} \Delta \rho] &= -\nabla \cdot b_\kappa - b_\kappa^\top S + \frac{\kappa}{2} (\nabla \cdot S + \|S\|^2) \\ &= -\frac{1}{2} \nabla \cdot R + \frac{1}{2} x^\top R - \frac{1}{2} \|R\|^2. \end{aligned}$$

Adding $g = (\|R\|^2 - Q)/2$ yields (4). Normalization subtracts $\mathbb{E}_{\rho/\int \rho g}$ from the potential. This calculation identifies the continuous path; existence of the normalized stochastic representation requires the corresponding integrability assumptions.

At $\kappa = 1$, this also specializes the weighted-product formula of Skreta et al. (2025, Proposition D.5): include the stationary factor $q_0 = \varphi$ with exponent $1 - G$ and each q_g with exponent one. The exponents sum to one, and the score-norm expression reduces to $(\|R\|^2 - Q)/2$. Other $\kappa > 0$ follow by adding $(\kappa - 1)\nabla \log \rho/2$ to the drift and changing the diffusion variance to κ ; their contributions cancel because $\nabla \cdot (\rho \nabla \log \rho) = \Delta \rho$.

B Gaussian tail envelopes under an Euler transition

We give the full scalar argument used by Theorem 3. Suppose

$$-\frac{x^2}{2V} - L|x| - C \leq \log p(x) \leq -\frac{x^2}{2V} + L|x| + C,$$

and $g(x) = cx^2 + r(x)$ with $|r(x)| \leq L_g(1 + |x|)$. Then

$$-\frac{D}{2V}x^2 - (L + hL_g)|x| - C' \leq \log[p(x)e^{hg(x)}] \leq -\frac{D}{2V}x^2 + (L + hL_g)|x| + C', \quad D = 1 - 2hcV.$$

For $D > 0$, the integral is positive and finite, and its normalization has the same quadratic envelope with $V_t = V/D$. For $D < 0$, the lower bound has a positive quadratic exponent and the integral diverges. At $D = 0$, the remaining terms can give either behavior, so the general statement excludes equality.

Now suppose the propagated state is

$$Y = AX + q(X) + \tau\xi, \quad |q(x)| \leq K, \quad \tau^2 = \kappa h > 0,$$

where X has the tilted density and $\xi \sim \mathcal{N}(0, 1)$ is independent. For $z = y - Ax$, the Gaussian transition-density ratio satisfies

$$e^{-K|z|/\tau^2 - K^2/(2\tau^2)} \leq \frac{\varphi_\tau(z - q(x))}{\varphi_\tau(z)} \leq e^{K|z|/\tau^2}.$$

Use the reference joint density $\varphi_{\sqrt{V_t}}(x)\varphi_\tau(y - Ax)$. Its Y marginal is Gaussian with variance $W = A^2V_t + \tau^2$, and its conditional distribution is

$$X | Y = y \sim \mathcal{N}\left(\frac{AV_t}{W}y, \frac{V_t\tau^2}{W}\right).$$

Multiplying the density-ratio bounds by the envelope for the tilted input reduces the output bounds to conditional expectations of $\exp\{\pm(L_1|X| + L_2|y - AX|)\}$. For fixed $L_1, L_2 \geq 0$, Gaussian exponential moments and the triangle inequality give an upper bound $e^{C_1|y| + C_2}$. Jensen's inequality gives the lower bound

$$\mathbb{E}[e^{-L_1|X| - L_2|y - AX|} | Y = y] \geq e^{-L_1\mathbb{E}[|X| | Y=y] - L_2\mathbb{E}[|y - AX| | Y=y]} \geq e^{-C_3|y| - C_4}.$$

It follows that $\log p_Y(y) = -y^2/(2W) + O(1 + |y|)$. The argument also covers $A = 0$ and bounded measurable q without a monotonicity assumption. On a fixed finite grid with a fixed finite batch size, the finitely many factors, batches, and histories admit common finite residual and envelope constants up to each strictly positive step. Every such conditional normalizer is finite and positive. Repeating the argument along these histories proves Theorem 3. For diagonal product models, a fixed index history leaves the coordinates conditionally independent, and the scalar argument applies coordinate by coordinate.

C A certificate for arbitrary affine controls

Suppose the true residual and its affine control have tail slopes $-a_g$ and $-t_g$, respectively. Set $e_g = a_g - t_g$, $T = \sum_g t_g$, and $c_0 = (T^2 - \sum_g t_g^2)/2$. The control-variate formula (16) remains conditionally unbiased when its residual is unbounded. For the empirical batch distribution p on the G factors, let $m = \sum_g p_g e_g$ and $s^2 = \sum_g p_g (e_g - m)^2$. Its quadratic coefficients are

$$\hat{A}(p) = T + Gm, \quad (18)$$

$$\hat{c}(p) = c_0 + GTm - G \sum_g p_g t_g e_g + \frac{1}{2}G(G-1)m^2 - \frac{1}{2} \left(\frac{G^2}{M-1} + G \right) s^2. \quad (19)$$

This identity follows by collecting the quadratic terms in (16); the bounded parts of mixture scores contribute only $O(1 + |x|)$ to the potential.

Proposition 4 (Constant-batch certificate). For Gaussian factors with arbitrary affine controls, the exact largest reachable variance is obtained by applying (11) to the G constant batches at each step and retaining the largest value. The corresponding integrability decision is independent of every finite $M \geq 2$. For equal-component-variance Gaussian mixture factors, the same recursion is exact for strict positive and strict negative denominator decisions under the tail-envelope assumptions of Theorem 3.

Proof. Delete the nonpositive variance term from (19), obtaining $\tilde{c}(p)$. This upper bound is convex on the probability simplex: its only quadratic term is the positive multiple $G(G-1)m^2/2$ of a squared affine function. Therefore $\tilde{c}(p) \leq \max_g \tilde{c}(\delta_g)$, and all vertex values are attained by constant batches. If a denominator fails at the largest reachable variance, it fails at a vertex.

On the common positive-denominator domain, replacing $\hat{c}$ by $\tilde{c}$ increases the next variance. Write

$$n(p) = \sqrt{V}[1 - h((1 + \kappa)(T + Gm) + \kappa)/2], \quad d(p) = 1 - 2hV\tilde{c}(p).$$

Here n is affine and d is positive and concave. The quadratic perspective

$$n(p)^2/d(p) = \sup_{z \in \mathbb{R}} \{2zn(p) - z^2d(p)\}$$

is convex in p , so its maximum on the simplex is attained at a vertex. The upper bound is exact there because $s^2 = 0$. Monotonicity in V and induction complete the Gaussian proof. The mixture extension follows from Appendix B, since every fixed batch still has an affine drift plus a bounded term and a quadratic potential plus a linearly bounded term. $\square$

For the constant batch containing factor g , the coefficients used by the implementation are

$$A_g^* = T + Ge_g, \quad c_g^* = c_0 + G(T - t_g)e_g + \frac{1}{2}G(G-1)e_g^2.$$

The check requires G candidates per step and has no combinatorial dependence on M . For exact tail slopes $t_g = a_g$, every $e_g = 0$, and all candidates coincide with the full-factor coefficients. With zero control, the simplex reduces to the scalar mean interval used in Theorem 2.

Corollary 5 (Paired correction preserves the strict certificate). Under the strict-domain conditions of Proposition 4, averaging two independent batch drifts and using the paired weight from Appendix D gives the same strict integrability decisions as the unpaired estimator. The statement also applies to the raw unbiased estimator.

Proof. The negative squared-difference correction can only reduce the weight. Omitting it gives quadratic potential coefficient $\bar{c} = (c_1 + c_2)/2$ and drift coefficient $\bar{A} = (A_1 + A_2)/2$. The variance expression

$$(A, c) \mapsto \frac{V[1 - h((1 + \kappa)A + \kappa)/2]^2}{1 - 2hVc} + \kappa h$$

is jointly convex in (A, c) on its positive-denominator domain, by the same quadratic-perspective identity. It is therefore at most the larger of the two unpaired next variances. Gaussian upper envelopes propagate under the omitted-correction transition; the actual positive weight obeys that upper bound after division by a finite positive normalizer. This proves finiteness along every paired history when the unpaired certificate stays strictly positive.

Conversely, take the maximizing constant batch at each step and use that same batch for both copies. This paired history has positive probability, the correction vanishes identically, and its tail variance follows the unpaired maximizing recursion. A strictly negative denominator therefore gives an infinite conditional integral. Tonelli’s theorem gives divergence of the full paired operator. Gaussian factors also allow the exact equality decision; the mixture result leaves equality unclassified. $\square$

D Second-order terms and paired weighting

This section records a local expansion to clarify the paired-correction ablation. Assume sufficient smoothness and integrable domination to justify expansion and normalization through third order for h in a neighborhood of zero. A bounded-state setup with bounded smooth coefficients is one sufficient setting. These assumptions must be checked before applying the expansion; it cannot justify a kernel with an infinite normalizer.

At a fixed time, suppose $\mathbb{E}_B \hat{b} = b$ and $\mathbb{E}_B \hat{g} = g$. Define $C_b(x) = \text{Cov}_B(\hat{b}(x))$, $c_{bg}(x) = \text{Cov}_B(\hat{b}(x), \hat{g}(x))$ and $v_g(x) = \text{Var}_B(\hat{g}(x))$. Let Q denote the random kernel and Q^0 the full-coefficient Euler kernel. Expansion of the unnormalized kernel yields

$$(Q - Q^0)f = h^2 \left\{ \frac{1}{2} C_b : \nabla^2 f + c_{bg}^\top \nabla f + \frac{1}{2} v_g f \right\} + O(h^3).$$

The diffusion-dependent terms are common to both kernels. Expanding the ratio in (7), the normalized population difference is

$$\Phi(\eta)(f) - \Phi^0(\eta)(f) = h^2 \left\{ \frac{1}{2} \eta(C_b : \nabla^2 f) + \eta(c_{bg}^\top \nabla f) + \frac{1}{2} \text{Cov}_\eta(f, v_g) \right\} + O(h^3). \quad (20)$$

The last term uses the covariance over the input population. A pointwise normalized kernel would be a different object. In particular, conditional score error alone does not specify this bias: its coupling with the potential enters through c_{bg} .

For two independent copies, use $\bar{b} = (\hat{b}_1 + \hat{b}_2)/2$ and weight

$$\exp \left\{ \frac{h}{2} (\hat{g}_1 + \hat{g}_2) - \frac{h^2}{8} (\hat{g}_1 - \hat{g}_2)^2 \right\}.$$

The expected squared difference is $2v_g$, cancelling the second-order potential-variance term. The remaining coefficient is

$$\frac{1}{4} \eta(C_b : \nabla^2 f) + \frac{1}{2} \eta(c_{bg}^\top \nabla f).$$

Thus paired weighting does not remove all second-order effects. For Gaussian factors, the event that both batches equal the same extreme batch at every step has positive probability; the correction vanishes on that history. Whenever Theorem 2 gives a divergent extreme history, this paired version also has an infinite normalizer.

E Without-replacement comparison

The with-replacement assumption is substantive. For a uniformly sampled subset of size $2 \leq M \leq G$, an unbiased potential estimator is

$$\hat{g}_{\text{WOR}} = \frac{G(G-1)}{M(M-1)} \sum_{i < j} r_{I_i}^\top r_{I_j}.$$

Distinct indices are sampled, so its pair coefficient differs from (6). For the same one-step example (13), exact subset enumeration gives the following maximum quadratic coefficients

and minimum denominators:

M	2	3	4
$\max \hat{c}_{\text{WOR}}$	2	29/33	346/693
$\min D$	47/355	659/1065	2503/3195

All three without-replacement one-step operators are finite. At $M = G$ the estimator equals the full-factor potential. For general grids, the following result provides an exact multi-step decision.

Theorem 6 (Prefix/suffix certificate). Assume scalar Gaussian factors, $a_g(u_k) \geq 0$, an initial Gaussian variance $V_0 > 0$, a finite grid, $\kappa > 0$, and independent uniform subsets of fixed size $2 \leq M \leq G$ at successive steps. At each level, sort the residual precisions as $a_1 \leq \dots \leq a_G$ and form

$$S_j = \{1, \dots, j\} \cup \{G - (M - j) + 1, \dots, G\}, \quad 0 \leq j \leq M.$$

For a subset S , define

$$\hat{A}_S = \frac{G}{M} \sum_{i \in S} a_i, \quad \hat{c}_S = \frac{G(G-1)}{M(M-1)} \sum_{\substack{i < \ell \\ i, \ell \in S}} a_i a_\ell.$$

Starting from $V_{\max,0} = V_0$, evaluate (10) only on these $M+1$ subsets at the current $V_{\max,k}$. If any denominator is nonpositive, the population normalizer is infinite at that step. Otherwise set

$$V_{\max,k+1} = \max_{0 \leq j \leq M} F_{S_j}(V_{\max,k}),$$

where F is (11). This recursion gives the largest reachable conditional variance and an exact integrability decision. Repeated strengths are allowed, with ties ordered by index.

Proof. Fix $M-1$ selected elements and vary the last strength z . Both $\hat{A}(z)$ and $\hat{c}(z)$ are affine in z , since each term in the distinct-pair sum contains z at most once. First maximize $\hat{c}$ over subsets. If a selected index lies strictly between the least and greatest unselected indices, replace it by whichever of those endpoints does not decrease $\hat{c}$. The interval between the least and greatest unselected indices strictly shrinks: the removed selected index is interior, and one unselected endpoint disappears. Iterating must terminate, and the remaining unselected indices then form a contiguous block. The selected set is therefore some S_j . Consequently a nonpositive denominator exists among all subsets if and only if it exists among these candidates.

Now suppose all candidate denominators are positive, hence all subset denominators are positive. With the other $M-1$ elements fixed, write the affine Euler multiplier as $n(z)$ and the positive affine denominator as $D(z)$. Between any two feasible endpoint strengths the denominator remains positive, and

$$F(z) - \kappa h = \frac{Vn(z)^2}{D(z)} = V \sup_{t \in \mathbb{R}} \{2tn(z) - t^2 D(z)\}$$

is convex as a supremum of affine functions of z . The same endpoint exchange can therefore be made without decreasing F . The strictly shrinking unselected interval again proves that a maximizing subset belongs to the $M+1$ candidates.

Finally, F_S is nondecreasing in V wherever its denominator is positive, because

$$\frac{d}{dV} \left(\frac{Vn_S^2}{1 - 2h\hat{c}_S V} \right) = \frac{n_S^2}{(1 - 2h\hat{c}_S V)^2} \geq 0.$$

Each finite history and each current subset have positive probability. A history attaining the preceding largest variance can thus be concatenated with a maximizing current subset. Induction proves attainability and maximality, while Lemma 1 and Tonelli's theorem prove the failure criterion. $\square$

Gaussian means and the linear potential terms do not affect this recursion by Lemma 1. For diagonal product factors, the coordinatewise criteria give an exact joint integrability decision: if a coordinate fails, its witnessing history makes the joint nonnegative integral infinite; if every coordinate passes, every history is integrable. The coordinate maxima need not occur on a common history. The same subset reduction supplies the strict tail-envelope certificate for the equal-component-variance mixture class of Appendix B, leaving zero-denominator mixture cases undecided.

For tail control without replacement, replace the last line of (16) by

$$\frac{G(G-1)}{M(M-1)} \sum_{\substack{i < j \\ i, j \in S}} e_i^\top e_j.$$

The preceding constant and linear estimators retain their forms with sums over S . Inclusion probabilities $\Pr(i \in S) = M/G$ and $\Pr(i, j \in S) = M(M-1)/(G(G-1))$ give conditional unbiasedness. Bounded residuals again make the random remainder grow at most linearly, so Theorem 3 applies to this sampling scheme as well. These statements concern the specified uniform subset estimator; they do not extend the prefix/suffix ordering argument to arbitrary affine controls.

F Complete benchmark certificate

Every executed benchmark configuration is classified in Figure 4. The coordinate-wise tests classify the population operator before inspecting any particle outcome.

Population certificate: F = finite, I = infinite, - = not run

Gaussian, G=16, K=128	F	F	I	I	I	I	F	F	-	-
Gaussian, G=16, K=512	F	F	I	I	I	I	F	F	-	-
Gaussian, G=16, K=2048	F	F	I	I	I	I	F	F	-	-
Gaussian, G=64, K=128	I	F	I	I	I	I	I	I	-	-
Gaussian, G=64, K=512	F	F	I	I	I	I	F	F	-	-
Gaussian, G=64, K=2048	F	F	I	I	I	I	F	F	-	-
Mixture, G=16, K=128	F	F	I	I	I	I	I	I	F	F
Mixture, G=16, K=512	F	F	I	I	I	I	I	I	F	F
Mixture, G=16, K=2048	F	F	I	I	I	I	I	I	F	F
Mixture, G=64, K=128	I	F	I	I	I	I	I	I	I	I
Mixture, G=64, K=512	F	F	I	I	I	I	I	I	F	F
Mixture, G=64, K=2048	F	F	I	I	I	I	I	I	F	F
Weak mixture, G=16, K=128	F	F	I	I	I	I	F	F	F	F
Weak mixture, G=16, K=512	F	F	F	F	F	F	F	F	F	F
Weak mixture, G=16, K=2048	F	F	F	F	F	F	F	F	F	F
Weak mixture, G=64, K=128	F	F	I	I	I	I	F	F	F	F
Weak mixture, G=64, K=512	F	F	I	I	I	I	F	F	F	F
Weak mixture, G=64, K=2048	F	F	I	I	I	I	F	F	F	F
	Full	Unweighted	Plug-in	Unbiased	Unbiased + pair	Moment CV	Moment CV + pair	Tail CV	Tail CV + pair	

Figure 4: Complete certificate for the executed benchmark methods. The statuses agree for $d = 1$ and $d = 8$, so the dimensions share a row. Each row therefore represents ten executed cells per method across both dimensions and five seeds. The classification is derived from the population recursion, independently of particle outcomes.

G Factor families and implementation details

For indices $g = 0, \dots, G-1$ and $j = 0, \dots, d-1$, set $\theta_{gj} = 2\pi(g + 0.37j)/G$. Each factor is a product over coordinates. The Gaussian family uses variance $v_{gj} = 0.5 + 0.15 \sin \theta_{gj}$

and mean $0.35 + 0.3 \cos \theta_{gj}$. The two-component mixture family uses the same component variance, component means

$$0.1 \cos \theta_{gj} \pm (0.7 + 0.2 \cos(2\theta_{gj})),$$

and positive-component weight $0.5 + 0.1 \sin(\theta_{gj} + 0.7)$. The weak mixture family uses

$$v_{gj} = \left[1 + \frac{4 + 2 \sin \theta_{gj}}{G} \right]^{-1}, \quad \mu_{gj,\pm} = \frac{0.4 \cos \theta_{gj}}{G} \pm \frac{0.6 + 0.1 \cos \theta_{gj}}{\sqrt{G}},$$

with the same component weights. All component variances are below one, so their residual tail precisions are nonnegative. The Gaussian and ordinary mixture have identical tail certificates despite their different interior geometry.

At each step, weights are multiplied using the left-endpoint potential and states receive the Euler drift and Gaussian noise. Systematic resampling is performed when the effective sample size is below half the particle count. Computation uses double precision, one GPU per task, four allocated CPU cores, and one numerical-library thread. Timing includes sampling, weighting, and resampling, with GPU synchronization at the measured boundaries. Factor reference quadrature and final metric computation are outside the sampler timing. The score-evaluation count records particle-factor evaluations and counts both batches in paired methods.

The raw endpoint sample for each benchmark cell contains all particles and normalized weights. Reference grids, densities, cumulative distribution functions, step metrics, runtime metadata, configuration, and completion records accompany each cell. The one-step study retains weighted histograms and the largest-weight particles, as specified before execution. Independent verification recomputes benchmark metrics using separate SciPy routines and checks all recovered file hashes against the remote manifest.

H Additional baseline comparisons

These tables retain the complete method comparisons summarized in Section 5.

Table 2: Number of configurations with finite population normalizers, out of 12 per family. A dash denotes an unrun method; tail matching coincides with the Gaussian control on Gaussian factors. Unweighted updates are Markov kernels with unit weights, so finiteness alone is immediate and does not establish target accuracy.

Method	Gaussian	Mixture	Weak mixture
Full	10/12	10/12	12/12
Unweighted	12/12	12/12	12/12
Plug-in	0/12	0/12	4/12
Unbiased	0/12	0/12	4/12
Unbiased + pair	0/12	0/12	4/12
Moment CV	10/12	0/12	12/12
Moment CV + pair	10/12	0/12	12/12
Tail CV	–	10/12	12/12
Tail CV + pair	–	10/12	12/12

I Extended sampling and learned-posterior protocols

The extension protocol is frozen separately from the 1,980-cell baseline. Its oracle matrix contains 1,080 unique sampling cells. For each of the three factor families, it uses $G = 64$, $d \in \{1, 8\}$, $K \in \{512, 2048\}$, five seeds, $P = 8192$ and $U = 20$. The methods are full weighting, unbiased weighting with and without replacement, and tail control with and without replacement. Each subsampled method uses $M \in \{2, 4, 8\}$; the full baseline is shared across batch sizes. This gives 780 cells. At $G = 64, d = 8, K = 512, M = 4$, the

Table 3: Matched configuration $G = 64, d = 8, K = 512$. W1 is mean $\pm$ one standard deviation over five seeds; seconds are medians. A superscript ∞ marks an infinite population normalizer. All methods use 8,192 particles.

Method	Mixture		Weak mixture	
	W1	Seconds	W1	Seconds
Full	0.881 ± 0.372	1.921	0.028 ± 0.007	1.909
Unweighted	0.404 ± 0.003	1.897	0.072 ± 0.001	1.896
Plug-in	$1.662 \pm 0.021^\infty$	0.822	$0.053 \pm 0.051^\infty$	0.754
Unbiased	$1.649 \pm 0.010^\infty$	0.828	$0.070 \pm 0.038^\infty$	0.764
Unbiased + pair	$1.608 \pm 0.099^\infty$	1.341	$0.032 \pm 0.014^\infty$	1.282
Moment CV	$0.742 \pm 0.097^\infty$	1.077	0.052 ± 0.040	1.047
Moment CV + pair	$0.456 \pm 0.228^\infty$	1.865	0.022 ± 0.005	1.850
Tail CV	1.004 ± 0.327	1.081	0.043 ± 0.026	1.027
Tail CV + pair	1.358 ± 0.188	1.876	0.025 ± 0.005	1.847

sensitivity study adds $P \in \{32768, 131072\}$ at $U = 20$ and $U \in \{10, 30\}$ at $P = 8192$ for every family, method and seed, adding 300 cells. Varying U with K fixed changes both initialization and the time grid, so this comparison does not isolate initialization error alone.

Without-replacement batches use independent double-precision uniform random keys for all G factors and select the M smallest keys. The resulting subset is uniform under the ideal continuous-key law. Selection requires $O(PG)$ work and is included in recorded sampler time. The displayed particle-factor evaluation count measures the selected scores and does not imply that subset selection is independent of G .

I.1 Conditional density learning

We use mixture density networks (Bishop, 1994) in a controlled neural-posterior setting; conditional density learning from simulator samples is established in simulation-based inference (Papamakarios & Murray, 2016). The simulator is

$$\theta \sim \mathcal{N}(0, 1), \quad Z \sim \text{Uniform}\{-1, +1\}, \quad y = \theta + Zb + \sigma\epsilon, \quad \epsilon \sim \mathcal{N}(0, 1).$$

During training, $\sigma \sim \text{Uniform}[0.45, 1.2]$ and $b \sim \text{Uniform}[0.3, 1.3]$. A network with three hidden layers of width 128 and SiLU activations maps $(y/3, \log \sigma, b)$ to four outputs. These parameterize two component means as center plus or minus a softplus separation, one common variance as $0.02 + 0.96 \text{sigmoid}(v)$, and two positive mixture weights using one logit. The variance restriction gives nonnegative residual tail precisions for the certificate. It is an architectural assumption, not an estimated property of a general score network.

Five independent training seeds each use 40,000 Adam updates with batches of 2,048 fresh joint simulator samples, giving 81,920,000 training pairs per seed. The learning rate decreases from 3×10^{-4} to 3×10^{-5} by a cosine schedule. Training minimizes conditional negative log likelihood and receives no exact posterior or score labels. We use the fixed final checkpoint for every seed. Validation on 100,000 fixed heldout pairs is recorded every 2,000 updates; it does not select checkpoints. Initial and final weights, loss histories and final checkpoint hashes are retained. A separate local audit evaluates the saved final networks on 100,000 newly generated heldout pairs and compares their negative log likelihood with the exact posterior value on those same pairs.

The exact conditional posterior, used only for evaluation, has common variance $v = \sigma^2/(1 + \sigma^2)$, means $(y - b)/(1 + \sigma^2)$ and $(y + b)/(1 + \sigma^2)$, and weights proportional to

$$\exp\left(-\frac{(y - b)^2}{2(1 + \sigma^2)}\right), \quad \exp\left(-\frac{(y + b)^2}{2(1 + \sigma^2)}\right).$$

An independent likelihood-times-prior calculation checks this identity. The learned factors are mixtures with a closed-form Ornstein–Uhlenbeck convolution, so their evolving scores and tail variances are available exactly conditional on the trained network. The experiment

therefore tests learned conditional densities with known tail structure, and does not certify arbitrary neural score models.

I.2 Grouped datasets and error decomposition

For each of five dataset seeds, $G \in \{16, 64\}$ and $d \in \{1, 8\}$, generate one shared $\theta_j \sim \mathcal{N}(0, 1)$ per coordinate and G conditionally independent observations. With $\alpha_{gj} = 2\pi(g + 0.37j)/G$, use

$$\sigma_{gj} = 0.45 + 0.75(0.5 + 0.5 \sin \alpha_{gj}), \quad b_{gj} = 0.8 + 0.5 \cos(2\alpha_{gj}).$$

Each observation has an independent sign and Gaussian noise. The same dataset and learned factor parameters are used by all five sampling methods. Every checkpoint/dataset pair is evaluated at $K \in \{512, 2048\}$ with $P = 32768$, $M = 4$, $U = 20$, $\kappa = 1$, standard Gaussian initialization, and the baseline resampling rule. This gives 1,000 sampling cells, across five training seeds, five dataset seeds, four (G, d) configurations, two grids and five methods.

We distinguish particle W1 against the normalized composition of learned factors, particle W1 against the true posterior composition, and the W1/KL discrepancy between the true and learned compositions. The two particle errors do not add to an exact decomposition of W1. Means and standard deviations across the 25 crossed checkpoint/dataset pairs describe dispersion; those pairs are not 25 independent training runs. The main figure uses the preselected finer grid, with the coarser grid retained in the appendix and complete tables. Network parameters are predicted once per context, before particle sampling. Setup time is recorded separately, and these timings do not measure repeated neural-network evaluations at evolving particle states.

I.3 Independent verification of the trained densities

All five networks contain 34,052 parameters and complete the prescribed 40,000 updates. Their initial and final checkpoints have finite weights and distinct hashes, with parameter-change Euclidean norms between 7.48 and 7.79. Table 4 reports every training seed. The independent conditional KL uses 1,024 new contexts and 16,385 quadrature points on $[-16, 16]$; both component-mixture densities integrate to one within the audit tolerance. The original GPU KL diagnostics use different heldout contexts and are retained separately.

Table 4: Fixed final checkpoints. Validation NLL uses the original 100,000-pair GPU validation set. Independent KL is $\mathbb{E}_c[\text{KL}(p(\theta | c) \| q_\psi(\theta | c))]$ averaged over 1,024 new contexts, with the displayed value multiplied by 10^5 . Every row is independently checked.

Training seed	Recorded validation NLL	Independent KL $\times 10^5$
0	1.091306	6.477
1	1.091242	4.828
2	1.091280	8.880
3	1.091315	9.242
4	1.091384	12.994

A separate 100,000-pair Monte Carlo comparison evaluates learned and exact-posterior negative log likelihood on the same fresh samples. Its excess-NLL estimates range from -2.38×10^{-5} to 1.72×10^{-4} , with standard errors between 3.29×10^{-5} and 5.13×10^{-5} . A negative finite-sample estimate is compatible with Monte Carlo noise; it is not a negative KL divergence. The integrated conditional KL and the composed-density errors supply separate distribution-level diagnostics.

J Complete extended sampling results

Figure 5 includes every oracle sampling configuration at $P = 8192, U = 20$. Across the full 1,080-cell oracle extension, all 600 full-factor or tail-controlled cells have finite normalizers;

all 480 raw unbiased cells have infinite normalizers. In particular, $M \in \{2, 4, 8\}$ remains below the finite thresholds at $G = 64$ for these tails. The statement is a certificate calculation, not an inference from observed particle errors.

J.1 Particle count and terminal-noise sensitivity

Figure 6 reports the full sweep at $G = 64, d = 8, K = 512, M = 4$. Increasing particles does not guarantee monotone improvement over this range. For the ordinary mixture, full-factor mean W1 decreases from 0.881 at $P = 8192$ to 0.650 at $P = 131072$, while tail control without replacement remains poor, with mean W1 1.440 and 1.470 respectively. For the weak mixture, raw with-replacement weighting improves from 0.0704 to 0.0098 while remaining nonintegrable, whereas finite tail control without replacement has mean W1 0.0250, 0.0194 and 0.0357 across the three particle counts. Thus increasing particles does not repair a failed population certificate, and the observed finite methods need not improve monotonically.

Changing U from 10 through 20 to 30 at fixed K leaves the full-factor and tail-control certificates finite and the raw certificates infinite in every tested cell. Full-factor mean W1 is 0.0265, 0.0280 and 0.0282 for the weak mixture, and 0.801, 0.881 and 0.711 for the ordinary mixture. Because this intervention changes the entire time grid as well as the initialization level, these data cannot identify an initialization-only effect.

J.2 Coarser learned-posterior grid

Figure 7 retains all 25 pairs at $K = 512$. Both grids have the same certificate classifications, while endpoint errors depend on the grid. Tables 5 and 6 report all configuration means and standard deviations; the released CSVs contain every individual cell, KS and moment diagnostics, prediction time and sampler time. The 100 distinct learned compositions are also retained individually.

Table 5: Complete learned-posterior results at $K = 512$. W1 values are means $\pm$ sample standard deviations across 25 crossed checkpoint/dataset pairs. Time is mean recorded sampler seconds. F and I denote finite and infinite population normalizers.

G	d	Method	Domain	W1 vs learned	W1 vs true	Seconds
16	1	Full	F	0.0207 ± 0.0116	0.0240 ± 0.0144	0.59
16	1	U-WR	I	0.0828 ± 0.0679	0.0824 ± 0.0690	0.76
16	1	U-WOR	I	0.0392 ± 0.0336	0.0417 ± 0.0350	0.73
16	1	Tail-WR	F	0.0420 ± 0.0391	0.0450 ± 0.0415	0.99
16	1	Tail-WOR	F	0.0339 ± 0.0223	0.0364 ± 0.0253	1.00
16	8	Full	F	0.0392 ± 0.0103	0.0408 ± 0.0111	1.94
16	8	U-WR	I	0.0907 ± 0.0238	0.0915 ± 0.0253	1.14
16	8	U-WOR	I	0.0740 ± 0.0253	0.0755 ± 0.0257	1.23
16	8	Tail-WR	F	0.0969 ± 0.0420	0.0978 ± 0.0416	1.26
16	8	Tail-WOR	F	0.0578 ± 0.0258	0.0598 ± 0.0258	1.36
64	1	Full	F	0.0102 ± 0.0057	0.0121 ± 0.0070	1.16
64	1	U-WR	I	0.0776 ± 0.0673	0.0779 ± 0.0671	0.79
64	1	U-WOR	I	0.0503 ± 0.0289	0.0508 ± 0.0285	0.80
64	1	Tail-WR	F	0.1336 ± 0.1299	0.1343 ± 0.1301	0.99
64	1	Tail-WOR	F	0.1130 ± 0.1096	0.1138 ± 0.1096	0.98
64	8	Full	F	0.0155 ± 0.0037	0.0173 ± 0.0051	6.56
64	8	U-WR	I	0.1365 ± 0.0488	0.1368 ± 0.0488	1.19
64	8	U-WOR	I	0.0807 ± 0.0180	0.0808 ± 0.0182	1.33
64	8	Tail-WR	F	0.2246 ± 0.0219	0.2248 ± 0.0206	1.33
64	8	Tail-WOR	F	0.1569 ± 0.0225	0.1573 ± 0.0226	1.48

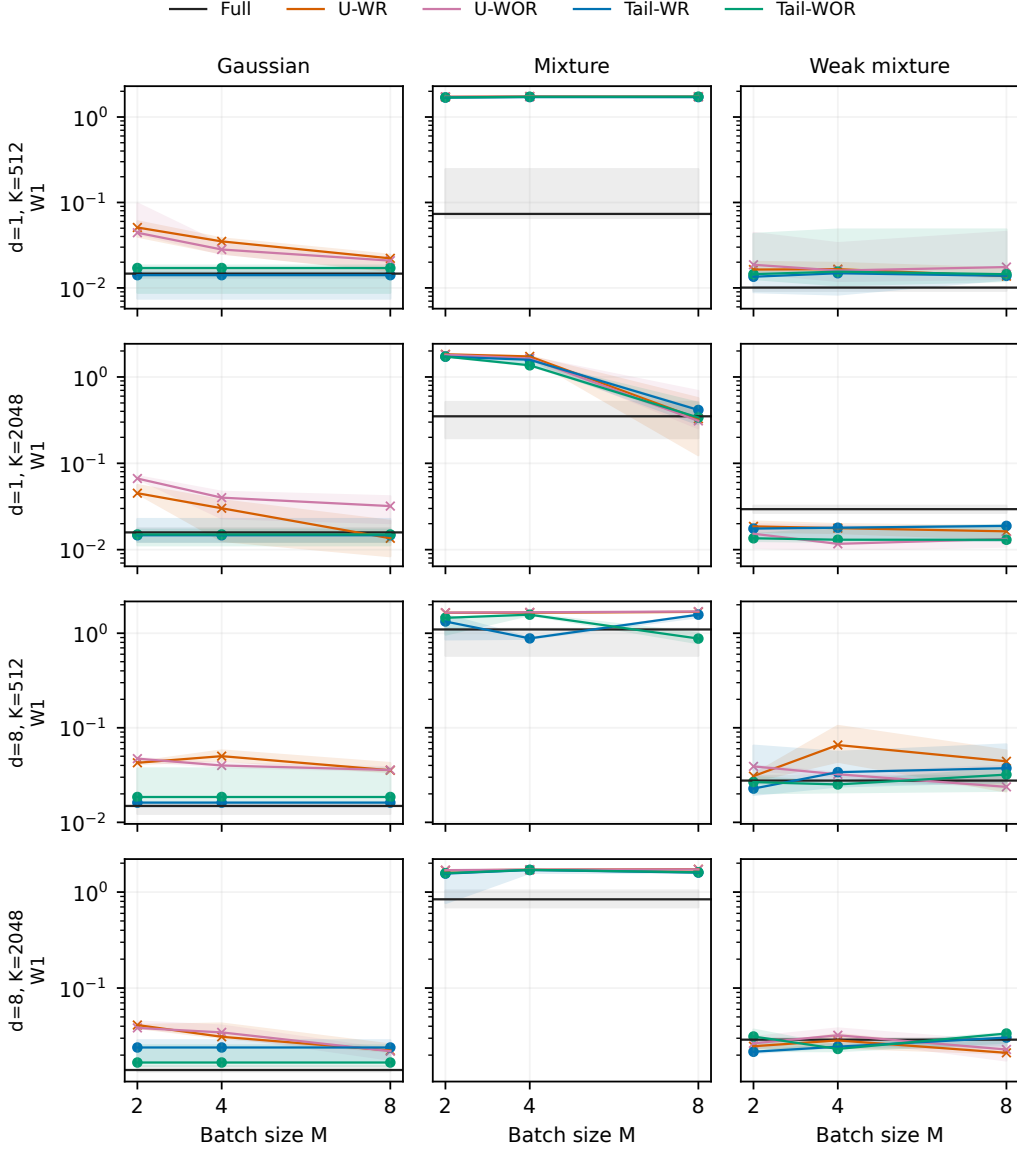


Figure 5: Complete oracle batch-size comparison. Curves show median marginal W1 and interquartile ranges across five seeds. The horizontal black line and band show the full-factor median and interquartile range. Circles denote finite normalizers and crosses infinite ones. Every row uses $G = 64, P = 8192, U = 20$; its dimension and grid size are labeled at left. All four subsampled methods and all three batch sizes are shown.

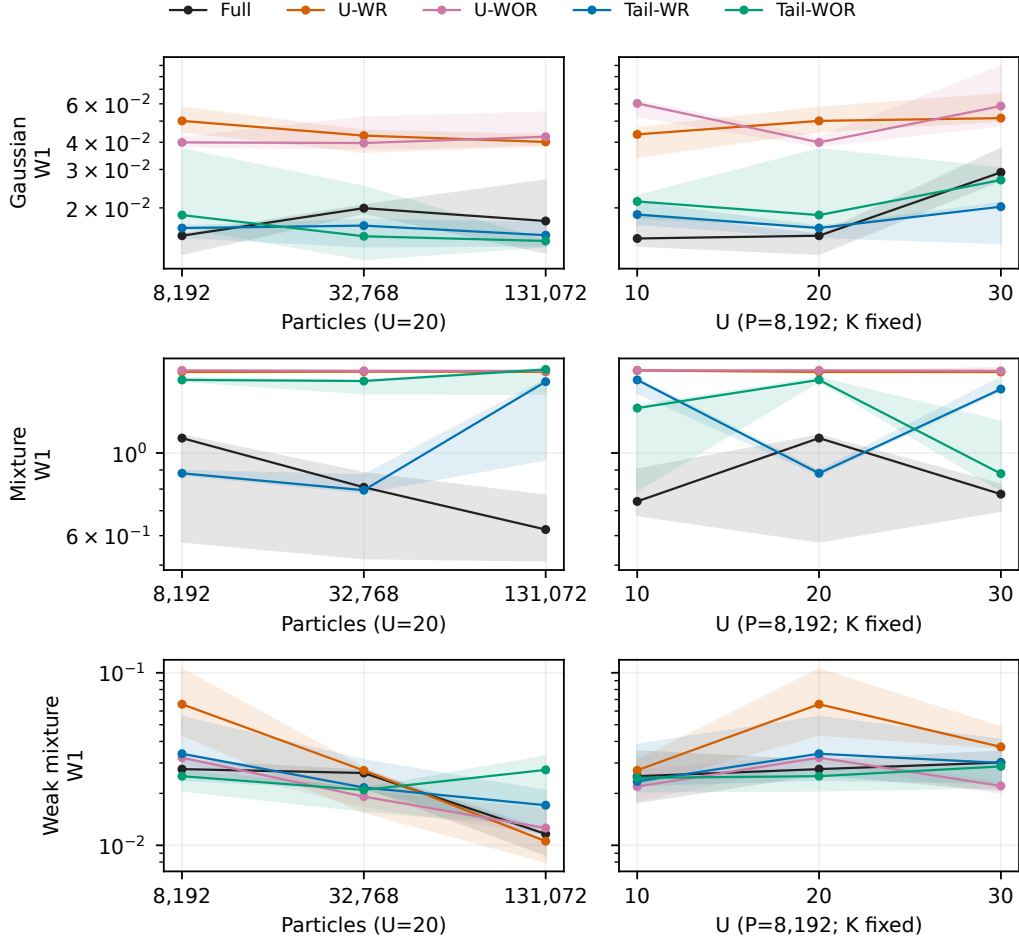


Figure 6: Complete particle-count and terminal-noise sweeps, with median marginal W1 and interquartile ranges across five seeds. Here circles mark every measured setting and do not encode validity: Full, Tail-WR and Tail-WOR are finite throughout; U-WR and U-WOR are infinite throughout. Left panels vary P at $U = 20$; right panels vary U at $P = 8192$. Both hold $K = 512$ fixed.

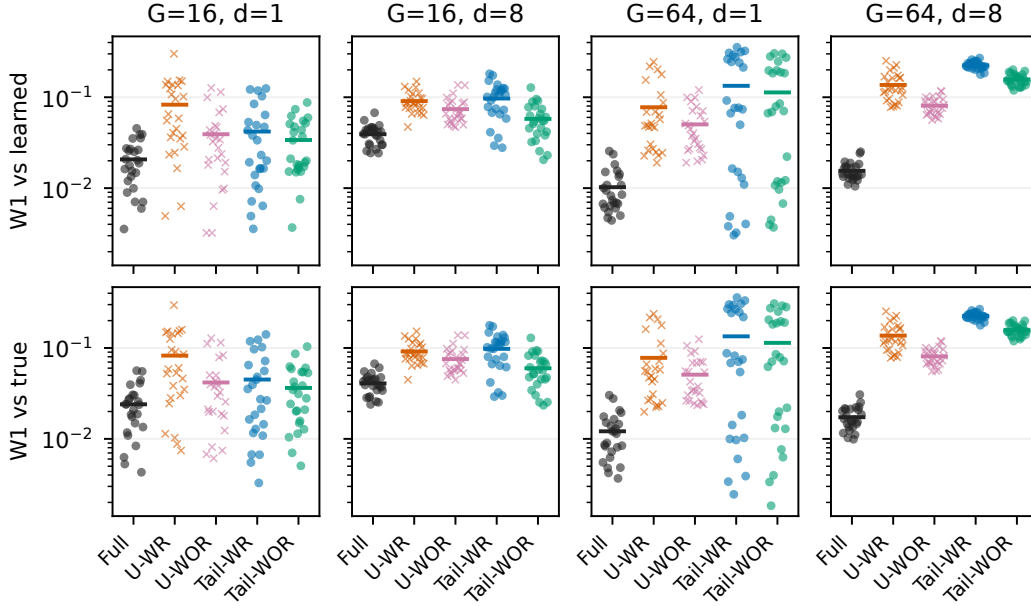


Figure 7: Complete learned-posterior comparison at $K = 512$, using the same display and validity conventions as Figure 3. All training and dataset seeds are included.

Table 6: Complete learned-posterior results at $K = 2048$. W1 values are means $\pm$ sample standard deviations across 25 crossed checkpoint/dataset pairs. Time is mean recorded sampler seconds. F and I denote finite and infinite population normalizers.

G	d	Method	Domain	W1 vs learned	W1 vs true	Seconds
16	1	Full	F	0.0192 ± 0.0108	0.0222 ± 0.0131	2.27
16	1	U-WR	I	0.0251 ± 0.0177	0.0275 ± 0.0175	2.80
16	1	U-WOR	I	0.0208 ± 0.0129	0.0248 ± 0.0143	2.72
16	1	Tail-WR	F	0.0256 ± 0.0306	0.0298 ± 0.0320	4.05
16	1	Tail-WOR	F	0.0208 ± 0.0136	0.0241 ± 0.0156	3.86
16	8	Full	F	0.0402 ± 0.0140	0.0424 ± 0.0157	7.70
16	8	U-WR	I	0.0572 ± 0.0253	0.0579 ± 0.0251	4.35
16	8	U-WOR	I	0.0400 ± 0.0130	0.0419 ± 0.0148	4.74
16	8	Tail-WR	F	0.0504 ± 0.0208	0.0517 ± 0.0213	4.94
16	8	Tail-WOR	F	0.0474 ± 0.0231	0.0486 ± 0.0242	5.38
64	1	Full	F	0.0125 ± 0.0080	0.0135 ± 0.0092	4.51
64	1	U-WR	I	0.0335 ± 0.0259	0.0344 ± 0.0278	3.07
64	1	U-WOR	I	0.0269 ± 0.0197	0.0291 ± 0.0193	2.91
64	1	Tail-WR	F	0.0610 ± 0.0586	0.0616 ± 0.0590	3.92
64	1	Tail-WOR	F	0.0487 ± 0.0402	0.0497 ± 0.0409	3.94
64	8	Full	F	0.0189 ± 0.0073	0.0201 ± 0.0073	26.09
64	8	U-WR	I	0.1006 ± 0.0337	0.1012 ± 0.0335	4.56
64	8	U-WOR	I	0.0448 ± 0.0130	0.0462 ± 0.0130	5.14
64	8	Tail-WR	F	0.1276 ± 0.0147	0.1279 ± 0.0151	5.13
64	8	Tail-WOR	F	0.0907 ± 0.0136	0.0912 ± 0.0129	5.73

K Frozen replication and tail-misspecification experiments

K.1 Protocol and uncertainty

A separately frozen protocol adds 630 sampling cells and 66 deterministic sensitivity configurations. All sampling cells use one NVIDIA H20, float64 particles, $G = 64$, $d = 8$, $U = 20$, $\kappa = 1$, the same quadratic time grid, and systematic resampling below an ESS fraction of 0.5. Configuration order is randomized before execution. Sampler times are synchronized GPU wall times for the sampling loop, including subset selection and diagnostic logging, and excluding reference integration, model prediction, certificate computation, and file packaging. Three preliminary smoke cells are excluded from all reported statistics.

The learned-density replication crosses the five fixed final checkpoints with 20 new dataset seeds, 100–119. At $K = 2048$, $P = 32768$, and $M = 4$, each of these 100 pairs is evaluated with full weighting, raw unbiased without-replacement weighting, and tail control without replacement, giving 300 cells. No checkpoint, dataset, or method is selected using the new results. We report mean marginal W1 against both the learned composition and the true posterior. Percentile intervals use 10,000 bootstrap draws, independently resampling the five training seeds and 20 dataset seeds on each draw while preserving method pairing. These intervals describe the limited training/data ensemble; the 100 pairs do not constitute 100 independent training replicates. The artifacts also retain the five per-training-seed means and intervals conditional on the fixed checkpoints.

The boundary study uses the three oracle families, $K \in \{512, 2048\}$, $P = 8192$, and ten new sampling seeds, 20–29. It compares full weighting, raw without-replacement weighting at $M_{\text{crit}} - 1$ and M_{crit} , and tail control at $M = 4$, giving 240 cells. Here M_{crit} is the previously reported first certified finite batch size, independently rechecked before sampling. The final 90 cells keep the target fixed and replace the component variance in the affine control by $\tilde{v}_g = (1 + \delta)v_g$, with $\delta \in \{-0.1, 0, 0.1\}$, $K = 512$, $P = 8192$, $M = 4$, and sampling with replacement. Control means are unchanged. This is a common multiplicative variance perturbation across factors and coordinates, not an arbitrary score-error model. Oracle intervals resample the ten sampling seeds with method pairing.

The deterministic sensitivity grid uses both time resolutions and

$$\delta \in \{-0.25, -0.1, -0.05, -0.025, -0.01, 0, 0.01, 0.025, 0.05, 0.1, 0.25\}.$$

Proposition 4 supplies the constant-batch certificate for these misspecified affine controls. The exact-tail case $\delta = 0$ agrees with full weighting in its quadratic coefficients. For nonzero δ , the estimator remains conditionally unbiased, but its strict integrability domain must be checked again.

K.2 Replication on new learned-density compositions

Full weighting and tail control are finite in all 100 new checkpoint/dataset pairs each; raw without-replacement weighting is infinite in all 100 pairs. Table 7 and Figure 8 report the complete comparison. The paired tail-minus-full W1 difference is 0.08042, with a crossed-bootstrap 95% interval of [0.07028, 0.09052]. The five training-seed means of this difference range from 0.07759 to 0.08390. Against the true posterior, the paired difference is 0.07929 with interval [0.06900, 0.08953]. The larger tail-control error therefore persists on the new datasets and across the existing training seeds.

Table 7: New learned-density replication: means and crossed-bootstrap 95% intervals over five fixed checkpoints and 20 new datasets. W1 is mean marginal Wasserstein distance. Times are mean H20 sampler seconds. The last column reports finite certificates out of 100 pairs.

Method	W1 to learned	W1 to true	Seconds	Finite
Full	0.01817 [0.01634, 0.02036]	0.01985 [0.01723, 0.02319]	14.06	100
Raw WOR	0.04725 [0.04252, 0.05240]	0.04776 [0.04308, 0.05285]	2.97	0
Tail WOR	0.09859 [0.08831, 0.10878]	0.09914 [0.08836, 0.10950]	3.40	100

Tail control takes about 24.2% of the sampler time of full weighting on this hardware and configuration, but its mean W1 to the learned composition is about 5.43 times as large. This is a fixed-particle, fixed-grid comparison, not a matched-accuracy speedup. All five networks are reused without further training, and their predictions are computed once per dataset; these timings do not represent repeated neural score evaluations.

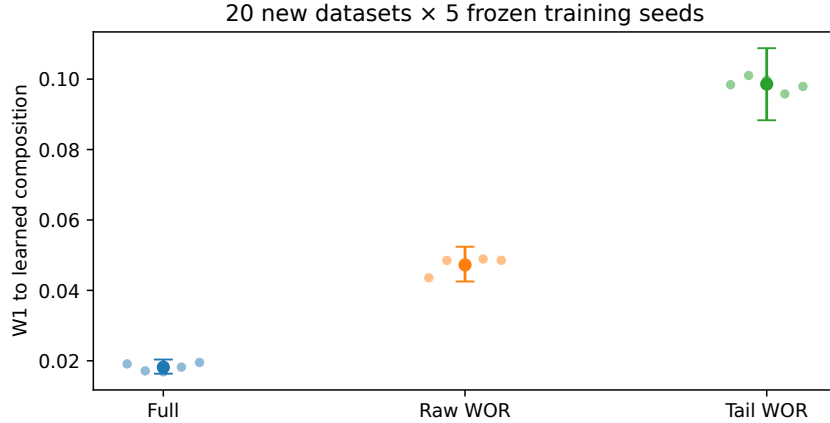


Figure 8: New learned-density replication. Large points show overall means with crossed-bootstrap 95% intervals; five smaller points show the means for individual training seeds, each averaged over the same 20 datasets. Raw WOR has an infinite population normalizer throughout, whereas Full and Tail WOR are finite throughout.

K.3 Sampling on both sides of a validity threshold

The rechecked critical batch sizes are 60 and 59 for the Gaussian and ordinary-mixture families at $K = 512$ and 2048, respectively, and 18 and 17 for the weak mixture. In each of these six settings, the certificate is infinite at $M_{\text{crit}} - 1$ and finite at M_{crit} . Both full weighting and $M = 4$ tail control are finite. Table 8 includes every setting, with intervals in Figure 9.

Table 8: Mean marginal W1 across ten new sampling seeds near the certificate boundary. Below and At use raw WOR with $M = M_{\text{crit}} - 1$ and $M = M_{\text{crit}}$, respectively. Below is infinite in every row; the other three conditions are finite. Tail uses $M = 4$.

Family	K	M_{crit}	Full	Below	At	Tail
Gaussian	512	60	0.01854	0.02385	0.01543	0.01911
Gaussian	2048	59	0.01965	0.01777	0.02172	0.02107
Mixture	512	60	0.86302	0.81897	0.68900	1.43500
Mixture	2048	59	0.85092	0.85712	0.66242	1.52086
Weak mixture	512	18	0.04455	0.03798	0.03498	0.03523
Weak mixture	2048	17	0.02948	0.03142	0.03945	0.03397

Observed accuracy changes in both directions across the validity boundary. For the ordinary mixture at $K = 2048$, the At-minus-Below difference is -0.19469 , with paired interval $[-0.33562, -0.05965]$. For the weak mixture at the same resolution, it is 0.00802 , with interval $[0.00061, 0.01784]$. These are descriptive intervals without adjustment across the six boundary comparisons. The certificate classifies existence of the population update; it does not require a monotone change in finite-particle endpoint error at the threshold.

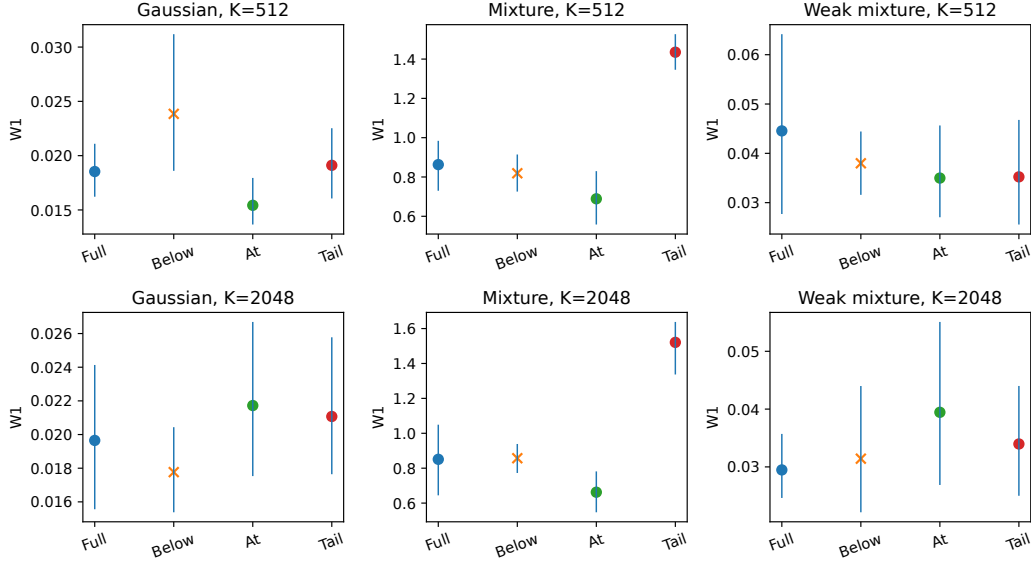


Figure 9: All six boundary comparisons, each with ten sampling seeds. Points show means and vertical bars show bootstrap 95% intervals. Circles denote finite normalizers and crosses infinite ones. Panel-specific vertical scales retain the differences in error magnitude between families.

K.4 Sensitivity to an imperfect tail control

Figure 10 reports all 66 deterministic perturbations. Every tested perturbation with $|\delta| \leq 0.1$ is finite for all three families and both grids. At $\delta = \pm 0.25$, the Gaussian and ordinary-mixture certificates are infinite on both grids, whereas the weak-mixture certificates remain finite. There are 58 finite and eight infinite configurations; no zero-denominator case is classified. These grid values demonstrate tolerance to the specified perturbations in the tested models, without establishing a uniform robustness interval for arbitrary factors or score errors.

Table 9: Mean marginal W1 under multiplicative control-variance error, with the target held fixed. Each entry averages ten seeds at $K = 512$, $P = 8192$, $M = 4$, with replacement. Every displayed setting has a finite certificate.

Family	$\delta = -0.1$	$\delta = 0$	$\delta = 0.1$
Gaussian	0.01974	0.01845	0.01882
Mixture	1.14131	1.18388	1.35780
Weak mixture	0.03086	0.03385	0.03260

Finite certificates do not eliminate sensitivity of sampling error. For the ordinary mixture, the $\delta = 0.1$ minus $\delta = 0$ difference is 0.17392, with a descriptive paired interval of $[0.00940, 0.36340]$. The complete seed-level results and all six perturbation-comparison intervals are retained, including intervals spanning zero.

Gaussian, K=512	∞	F	F	F	F	F	F	F	F	F	∞
Gaussian, K=2048	∞	F	F	F	F	F	F	F	F	F	∞
Mixture, K=512	∞	F	F	F	F	F	F	F	F	F	∞
Mixture, K=2048	∞	F	F	F	F	F	F	F	F	F	∞
Weak mixture, K=512	F	F	F	F	F	F	F	F	F	F	F
Weak mixture, K=2048	F	F	F	F	F	F	F	F	F	F	F
	-25.0%	-10.0%	-5.0%	-2.5%	-1.0%	+0.0%	+1.0%	+2.5%	+5.0%	+10.0%	+25.0%
Relative error in control component variance											

Figure 10: Strict-domain certificates for every tested control-variance perturbation. F denotes a finite normalizer and ∞ an infinite one. Columns are the discrete tested values of δ , not a continuous parameter grid or an assertion about untested values.

K.5 Independent verification and remaining scope

All 630 sampling cells completed with a zero launcher exit status, and every original particle array is retained. We reload all five checkpoints, regenerate the 20 datasets, and verify the 100 learned parameter sets and their consistency across methods. Independent SciPy reference calculations use 131,073 points on $[-16, 16]$, with a separately implemented Gaussian-mixture density. These checks cover 103 learned/oracle parameter settings and their corresponding true-posterior references. Independent recomputation checks W1, KS, means, variances, and positive-coordinate masses for every cell. The maximum W1 discrepancy against the original records is 1.61×10^{-6} . A separate vectorized recursion checks 333 sampling-certificate configurations and all 66 sensitivity configurations, including failure steps. Source hashes, checkpoint hashes, complete cell manifests, and the recovered raw-file hashes are retained.

This replication strengthens evidence within the controlled simulator and explicit Gaussian-tail model class. It introduces no additional training replicates and provides no validation on a real scientific inverse problem, arbitrary neural scores, or non-diagonal tails. The new data support the distinction between population validity and finite-particle accuracy; they do not establish general speed-accuracy superiority for tail control.